

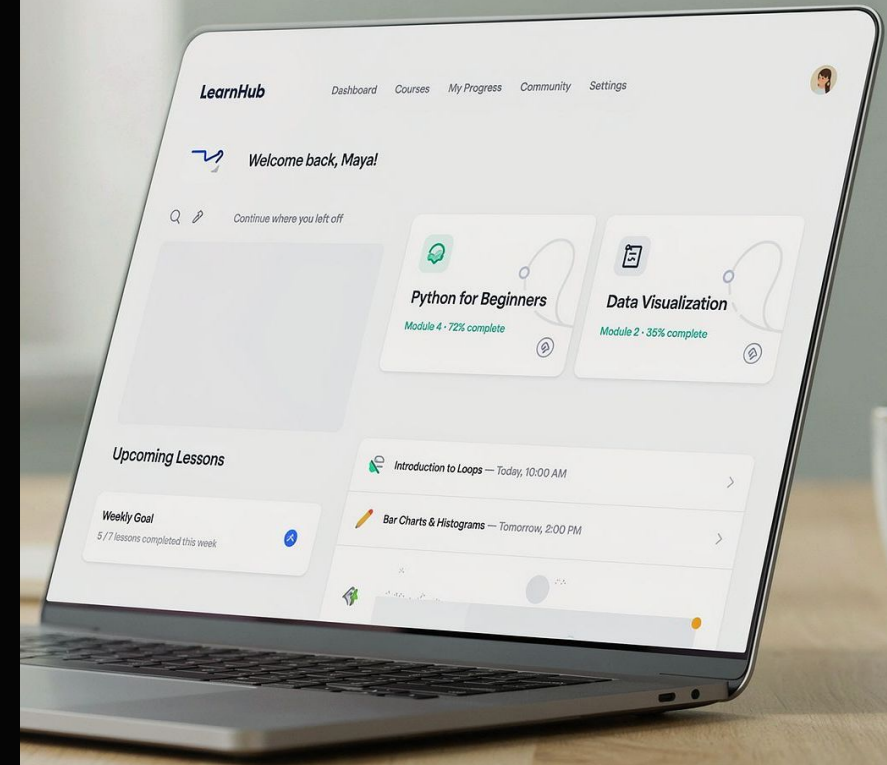
LMS Complete Feature Blueprint

A comprehensive, production-ready Learning Management System designed as two connected platforms — **User/Student** and **Admin/Management** — with an optional **Instructor** role. This blueprint covers all 40+ modules across five development phases, from MVP to AI-powered features.

FULL STACK ARCHITECTURE

40+ MODULES

5 PHASES



Two Connected Systems

The LMS is architected as two distinct but deeply integrated platforms. Each serves a different user class, yet shares the same underlying data and logic layer.

User / Student Platform

Course discovery, enrolment, video learning, progress tracking, quizzes, certificates, payments, wishlist, reviews, Q&A, and notifications — all in a clean, mobile-responsive interface.

Admin / Management Platform

Full control over users, courses, content, payments, analytics, marketing, permissions, and CMS — a powerful back-office that drives the entire platform.

Instructor Role (Optional)

A third role layer that lets approved instructors create and manage their own courses, view student progress, respond to Q&A, and track earnings — without full admin access.

User Side — Authentication & Account

Every student journey starts with a secure, friction-free account system. Support for multiple sign-in methods ensures maximum reach across audiences.

Registration & Login

- Email/password registration and login
- Google, Facebook, and Apple OAuth
- Phone / OTP verification
- Email verification on sign-up
- Forgot password and reset flow
- Remember me and session management

Profile Management

- Profile photo, name, bio
- Phone, email, location
- Social links (LinkedIn, GitHub, etc.)
- Change password
- Account deletion with confirmation
- 2FA for security-conscious users

Student Dashboard

The student dashboard is the command centre for every learner. It surfaces the right content at the right time, encouraging continued engagement and course completion.

Continue Learning

Resume the last-watched lesson instantly with progress bar, completion %, and a direct "Continue →" CTA.

My Courses

All enrolled courses — in-progress, completed, and recently viewed — organised in a scannable grid.

Certificates

Download or share earned certificates, each with a unique Certificate ID and verification link.

Wishlist & History

Saved courses, purchase history, payment receipts, and personalised course recommendations.

Learning Statistics

Total learning hours, streaks, quiz scores, and assignment statuses — all in a single view.

Notifications

Real-time alerts for new courses, replies, coupon drops, and course announcements.

Course Discovery & Search

A powerful browsing and search system is the gateway between learners and the right course. Think Udemy-level discoverability — every filter, sort, and keyword combination should surface accurate results quickly.

Browse & Listings

- All, new, popular, featured, best sellers
- Free, paid, discounted, recommended
- Search by name, instructor, category, tag, skill, keyword

Filters

- Category & subcategory
- Price range, rating, difficulty, duration
- Language, level, certificate availability

Sorting Options

- Most popular / highest rated
- Newest / lowest price / highest price
- Most enrolled

Search Experience

Full-text search with instant suggestions across course name, instructor, category, topic, and tags. Results update in real time as filters are applied.

Course Details Page

The course details page is one of the most conversion-critical pages on the platform. It must answer every question a prospective student has before they decide to enrol or purchase.

Course Identity

Title, subtitle, description, thumbnail, preview video/trailer, instructor name, category, subcategory, tags, level, language, duration, and lecture count.

Learning Outcomes

"What you'll learn" checklist, prerequisites/requirements, target audience, and clearly stated learning goals.

Course Curriculum

Expandable section/lesson tree with lesson type indicators (video, quiz, assignment), estimated duration per section, and free preview lessons.

Social Proof & Pricing

Star rating (e.g. ★ 4.8), student count, review excerpts, enrolment count. Prominent BUY NOW and ADD TO CART CTAs with coupon field and final price breakdown.



CHAPTER 3

Video Learning System

Video delivery is the core of your LMS. A great player experience combined with proper video security ensures students stay engaged and your content stays protected.

Player Features

- Play/pause, volume, fullscreen
- Playback speed (0.5x – 2x)
- Quality selection and captions/subtitles
- Auto-play next lesson
- Resume from last position
- Video progress tracking
- Picture-in-picture and keyboard shortcuts

Video Security (Critical)

Never expose raw video files publicly. Use:

- **Private storage** (AWS S3, Cloudflare R2)
- **Signed / expiring URLs** per request
- **Server-side access authorisation** before serving
- **Optional DRM** for premium content
- Streaming-only delivery — no direct download links

Course Structure & Content Types

A well-structured course hierarchy is essential for both content organisation and granular progress tracking. Every lesson can contain multiple content types to support diverse learning styles.

Hierarchy

01

Course

Top-level container with metadata, pricing, and enrolment settings.

02

Section

Grouped chapters within a course, each with a title and optional introduction.

03

Lesson

Individual learning units — the atomic content item tracked for completion.

Lesson Content Types

- Video (primary format)
- PDF and downloadable files
- Audio, text/rich content
- Quiz and assignment
- External resource / link
- Coding exercise (sandboxed)

Learning Progress Tracking

Granular progress tracking is what separates a serious LMS from a simple video hosting platform. Every interaction a student has with a lesson should be recorded and surfaced back to them meaningfully.

Per-Lesson Tracking

- Started / completed status
- Watch percentage
- Last watched position (resume)

Per-Course Metrics

- Overall completion %
- Sections completed
- Continue learning CTA

Platform-Wide Stats

- Total learning hours
- Learning streak (daily)
- Quiz scores history
- Assignment statuses

Quiz System

Quizzes reinforce learning and give both students and admins measurable outcomes. The system must be flexible enough for simple knowledge checks and complex timed assessments alike.

Question Types

- Multiple choice (single answer)
- Multiple answer (checkboxes)
- True / False
- Fill in the blank
- Matching pairs
- Randomised question order

Quiz Configuration

- Number of questions, passing %
- Time limit and attempts allowed
- Show answers after submission
- Negative marking option
- Per-question explanations

Student Result View

Score %, correct vs. wrong count, pass/fail badge, and detailed answer review — all presented immediately after submission.

Certificate System

Certificates are a major motivation driver for learners. Automatic generation, unique IDs, and public verification make your certificates credible and shareable.



Automatic Generation

A certificate is generated the moment a student completes all required lessons and quizzes in a course. No manual admin action needed.



Verification & Sharing

Each certificate has a unique Certificate ID, a public verification URL, and an embedded QR code. Students can download as PDF or share directly to LinkedIn.



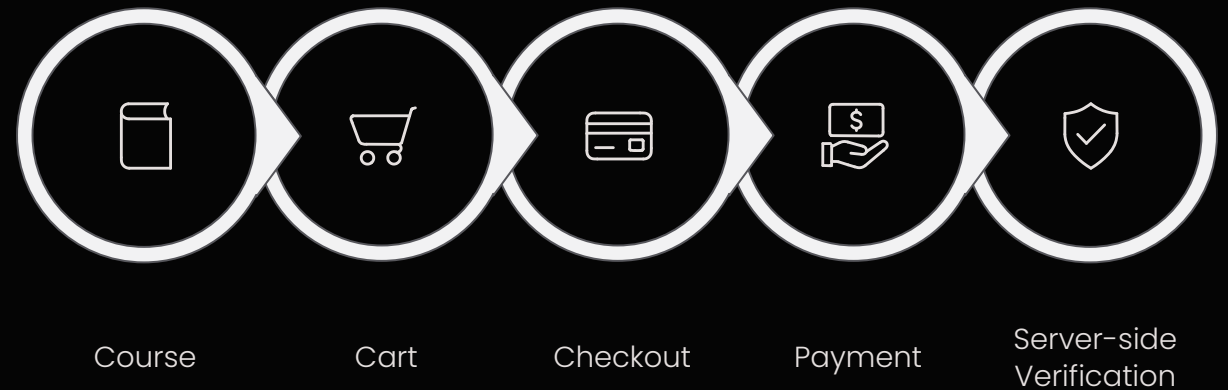
Template Management

Admins can create and manage multiple certificate templates – customising logo, colours, signatures, and layout for different course types or branding needs.

CHAPTER 5

E-Commerce & Payments

A robust purchase flow is non-negotiable for a paid LMS. The key rule: **never grant course access solely because the frontend says payment succeeded** — always verify server-side via the gateway's webhook or API.



Support Razorpay (India-first), Stripe, and PayPal. Collect billing info, apply coupons, calculate GST/tax, and display a clear order summary before the student ever hits the payment gateway.

Coupons, Discounts & Wishlist

Marketing and retention tools that drive conversions and keep students engaged with courses they haven't purchased yet.

Coupon Types (Admin Creates)

- Percentage discount (e.g. WELCOME20 → 20% off)
- Fixed amount discount
- Course-specific, category, or global scope
- Expiry date and usage limit
- Per-user usage limit
- Minimum purchase amount threshold

Wishlist Features

- Add / remove courses from wishlist
- Dedicated wishlist page
- Notify user when a wishlisted course goes on sale
- "Course price dropped" push/email notification

📌 Wishlist-to-purchase notifications are one of the highest-converting re-engagement tools in e-commerce. Don't skip them.

Ratings, Reviews & Q&A

Social proof and peer-to-peer learning are what make a marketplace-style LMS feel alive. Both systems need moderation controls to stay healthy.

★ Ratings & Reviews

Students give 1–5 stars and write reviews. They can edit or delete their own review. Rating distribution (5★ 85%, 4★ 10%...) is shown on the course page. Admin can hide or delete inappropriate reviews and moderate reported ones.

💬 Lesson Q&A

Students post questions under individual lessons. Instructors (or other students) reply in threaded discussions. Features include likes, "accepted answer" marking, instructor badge, report content, and reply notifications.

Notifications & Email System

A centralised notification system ensures students and admins never miss critical events — from new course availability to payment confirmations to instructor responses.

Notification Triggers

- Course purchased / payment successful
- Course completed / certificate generated
- Quiz result available
- Instructor replied to Q&A
- Coupon / price drop on wishlist course
- New course announcement

Delivery Channels

- In-app notifications (bell icon)
- Email notifications (transactional)
- Push notifications (browser/mobile)

Admin-Managed Email Templates

- Welcome, verification, password reset
- Purchase confirmation, payment failure
- Enrolment, certificate, promotional

Admin Dashboard

The admin dashboard is mission control for the entire platform. It surfaces the most critical platform health indicators at a glance, enabling fast, data-driven decisions.

25K+

Total Users

Students, instructors, and admins across the platform.

486

Total Courses

Published, draft, and pending approval.

₹48L+

Total Sales

Gross revenue across all payment gateways.

₹42L+

Net Revenue

After refunds, discounts, and gateway fees.

The dashboard also shows today's sales, new users, active users, new courses, plus visual graphs for sales trends, user growth, and course enrolment over time.



User Management

Admins need granular control over every user on the platform — from viewing purchase history to blocking bad actors.

User List & Search

View, search, filter, and sort all users by name, email, role, status, and join date. Paginated list with bulk actions.

Per-User Actions

- Create, edit, block, unblock, delete
- Reset password / verify email / verify phone
- Change role (student → instructor, etc.)
- View purchases, enrolled courses, progress
- View certificates, payments, and activity log

User Profile View (Admin)

A full snapshot: courses enrolled (7), completed (3), total spent (₹8,500), current status, and a timestamped activity feed.

Instructor Management

A true Udemy-like LMS decentralises course creation to instructors. The admin's job is to approve, monitor, and compensate them fairly — while maintaining full platform control.



Instructor Controls

Application, approval/rejection, suspension, verification, commission rate setting, and payout management.

Instructor Dashboard

Course creation, lesson upload, quiz creation, student view, earnings, course analytics, Q&A responses, and review replies.

Payout System

Gross sales → platform commission deducted → instructor net earnings → scheduled payout. Admin sees pending, processing, and paid balances.

Course Management — Admin

Full CRUD over every course on the platform. When instructors can create courses, admin becomes the editorial gatekeeper — reviewing and approving before anything goes live.

Course States

- **Draft** — being authored, not visible publicly
- **Pending Approval** — submitted by instructor, awaiting review
- **Published** — live and purchaseable
- **Unpublished** — temporarily hidden
- **Rejected** — returned to instructor with reason
- **Archived** — retired, not discoverable

Admin Course Actions

- Create / edit / delete any course
- Publish, unpublish, or archive
- Approve or reject instructor-submitted courses
- Manage sections, lessons, quizzes inline
- Override pricing, tags, and category
- View per-course analytics and reviews

Category & CMS Management

A flexible taxonomy system keeps the course catalogue organised. The CMS layer ensures the platform's public-facing content is always up to date without requiring a developer.

Category Management

- Hierarchical: Category → Subcategory
- Full CRUD — create, edit, delete, reorder
- Category image and banner
- SEO slug, title, and meta description

Example: Programming → Web Development → React

CMS — Pages & Banners

- Manage About Us, Privacy, Terms, Refund Policy, FAQ
- Rich text editor for all pages
- Homepage banners: image, title, subtitle, button URL, start/end date, display order
- Media library: upload, manage, and reference all platform images
- Promotional banners, course thumbnails, instructor images, logos, favicons

Homepage Management — Full CMS

Your homepage should never be hard-coded. Admin controls every section, its order, content, and visibility — turning the homepage into a true CMS-driven page.

1

Hero Banner

Full-width banner with headline, sub-headline, CTA button, and scheduled start/end dates.

2

Featured Sections

Categories, popular courses, best sellers, new arrivals — each section is toggleable and reorderable.

3

Social Proof

Instructor spotlight, student testimonials, and a promotional banner — all content managed from admin.

4

Footer & FAQ

FAQ section and footer links are managed via admin — no code deploys needed to update them.

Payments, Orders & Refunds

The financial backbone of the platform. Every transaction must be fully auditable, and refunds must follow a structured approval workflow.

Order & Payment Management

- All orders with item breakdown, discount, tax, and total
- Payment status: successful, failed, pending
- Gateway details (Razorpay, Stripe, PayPal)
- Invoice generation and download
- Failed payment retry tracking
- Tax/GST settings per gateway

Refund Workflow

01

Requested

Student submits refund request with reason.

02

Admin Review

Approved or rejected with an admin note.

03

Processed

Refund initiated via gateway; status moves to Completed.

Analytics — Users, Courses & Revenue

Data-driven decision-making requires comprehensive analytics across every dimension of the platform. These dashboards help identify what's working and where students are dropping off.



User Analytics

Total, new, active, and returning users. Retention rates, user growth over time, and geographic distribution.



Course Analytics

Enrolments, completion rate (e.g. 68%), average rating, per-lesson drop-off points, daily/monthly enrolment trends.



Revenue Analytics

Daily, weekly, monthly, and yearly revenue. Course revenue, instructor revenue share, refunds, taxes, and discount impact.

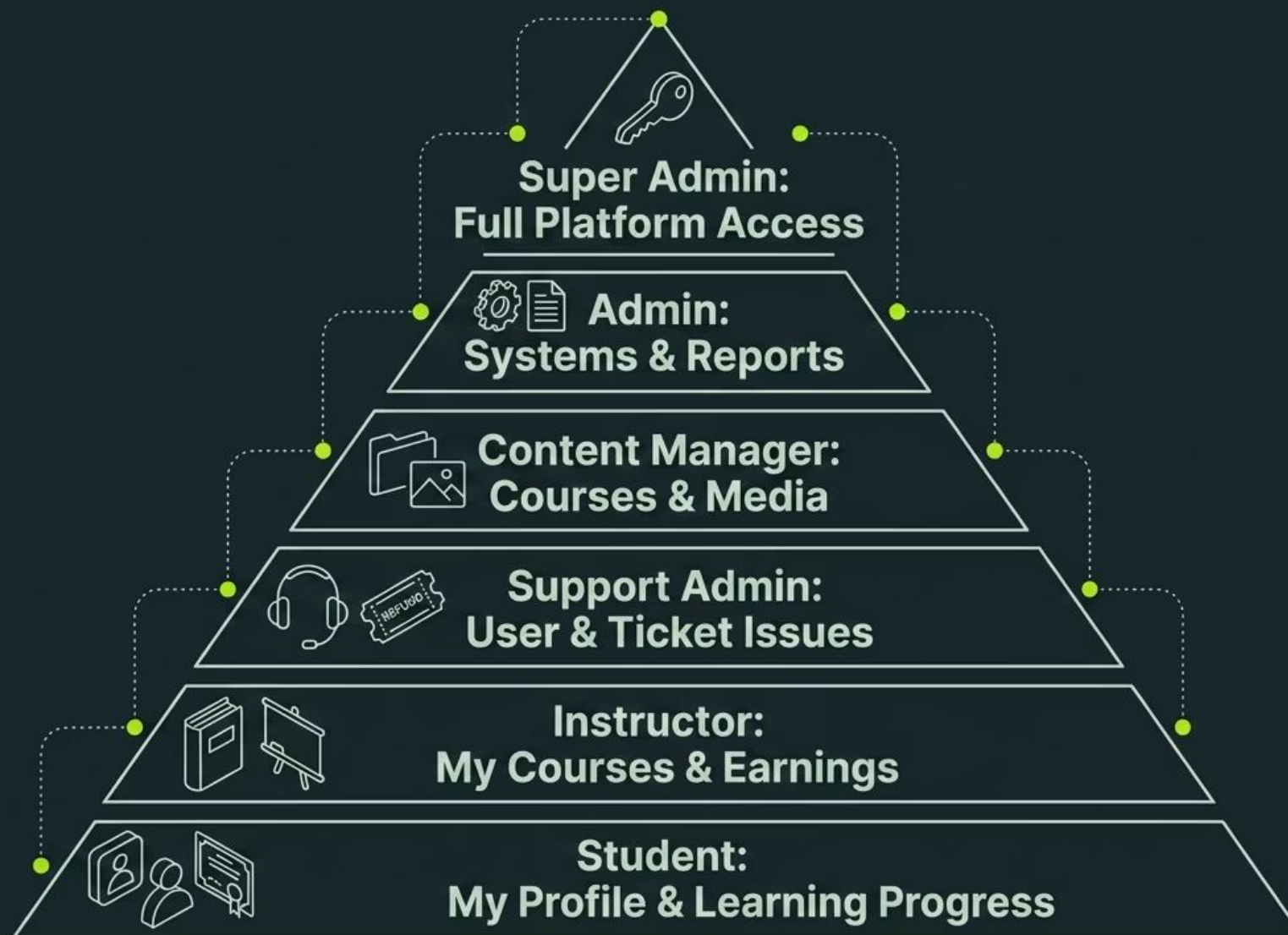


Learning Analytics

Total platform learning hours, overall completion rate, average course completion, quiz performance, and assignment submission rates.

Role & Permission System

Never reduce your authorisation to `admin = true`. A granular, role-based permission system is one of the most important architectural decisions you'll make — and one of the hardest to retrofit later.



Each permission is structured as `resource.action` — e.g. `course.publish`, `user.block`, `banner.delete`. Assign permission sets to roles, then assign roles to users.

Security — Non-Negotiable Foundations

Security is a first-class feature, not an afterthought. These protections must be implemented from day one, particularly around authentication, payments, and content delivery.

Application Security

- Password hashing (bcrypt/argon2)
- JWT / session security with refresh tokens
- CSRF protection on state-changing endpoints
- Rate limiting on auth and payment routes
- Input validation and sanitisation
- SQL injection & XSS protection
- Secure file upload with MIME validation

Admin & Content Protection

- 2FA mandatory for admin accounts
- Login activity and session management
- Full admin audit logs
- HTTPS everywhere, secure cookies
- Strict protection on `/admin`, `/api/admin/*`, `/payment/*`, `/course/private/*`
- RBAC + permission-based route guards

Support, Moderation & Audit Logs

A professional platform needs structured channels for user support, content moderation, and a tamper-evident record of every admin action.

1

Support Tickets

Students create tickets with subject, message, and attachments. Admin can reply, assign, close, or escalate. Categories: Payment, Technical, Content, Account.

2

Report & Moderation

Users report courses, reviews, questions, answers, or instructors. Admin sees a moderation queue with Pending / Reviewed / Dismissed statuses and can delete or take action.

3

Audit Logs

Every admin action is logged: who did what, to which resource, what changed, from which IP, and at what timestamp. Invaluable for security investigations and compliance.

SEO, Blog & Marketing System

For a public LMS, organic traffic is your most cost-effective acquisition channel. SEO, a blog, and an integrated marketing toolset work together to attract and convert learners at scale.

SEO Fundamentals

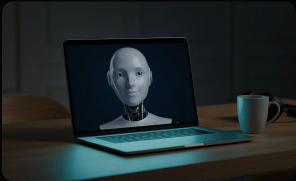
- SEO title, meta description, canonical URL per page
- Clean slugs: `/courses/full-stack-web-development`
- Open Graph and Twitter/X card tags
- Sitemap.xml and robots.txt
- Course schema structured data, breadcrumbs

Blog & Marketing Tools

- Blog posts with categories, tags, author, SEO, and scheduled publishing
- Admin-managed coupons, flash sales, email campaigns, push campaigns
- Promotional banners and landing pages
- Referral / affiliate system with commission tracking
- Affiliate links, conversions, earnings, and payout management

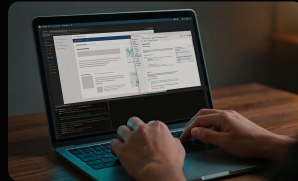
Advanced Features — Phase 5

Once the core platform is stable and generating revenue, these advanced capabilities transform your LMS from a course marketplace into an intelligent, engaging learning ecosystem.



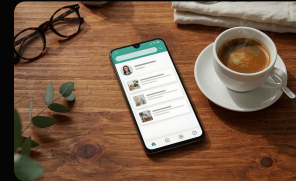
AI Features

- AI course assistant (ask questions while watching)
- AI quiz generation from lesson content
- AI-powered course recommendations



Coding Sandbox

- In-browser code editor and execution
- Multi-language support with test cases
- Automatic grading and project submissions



Mobile App

- Android and iOS native apps
- Offline video access (encrypted)
- Push notifications, payments, certificates

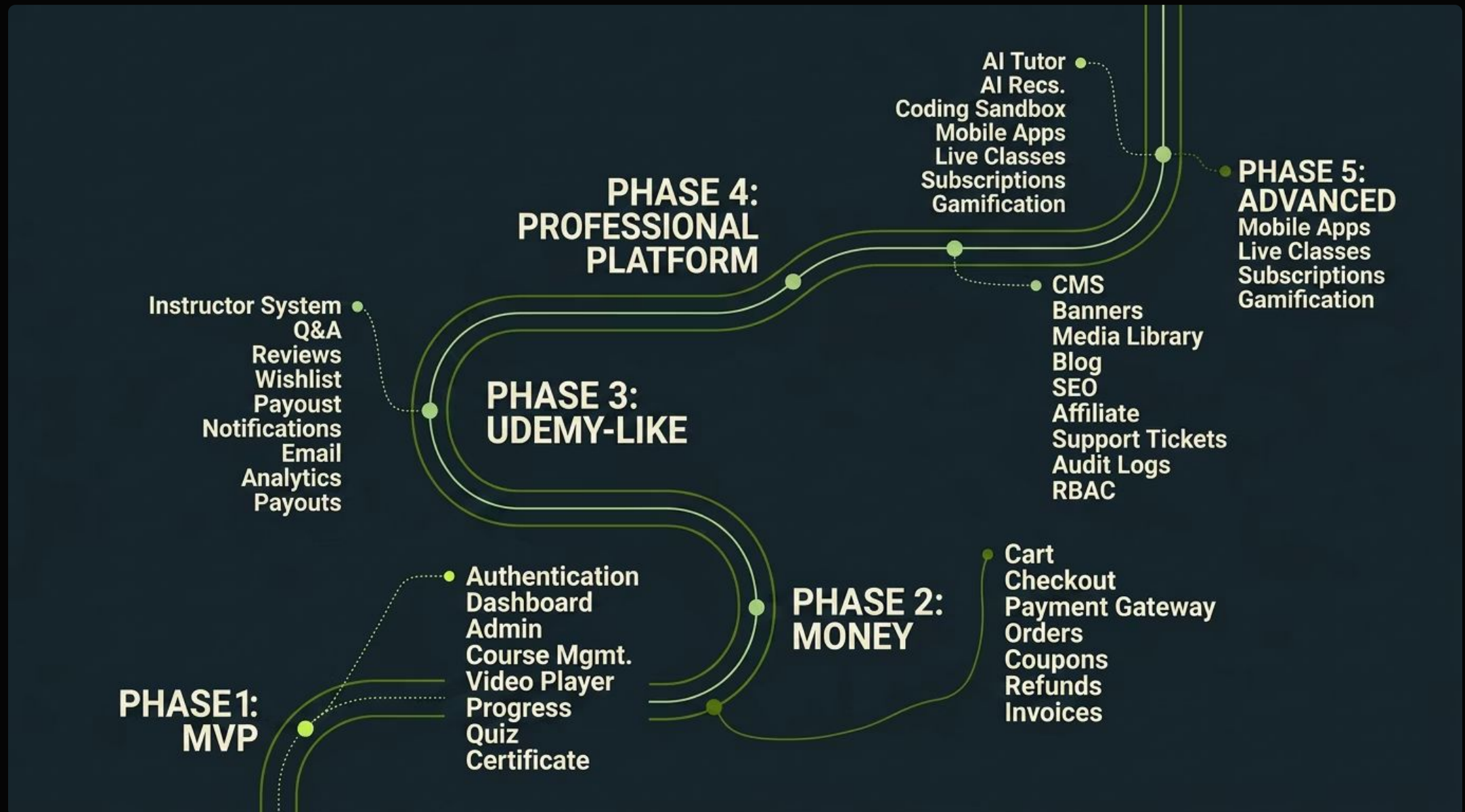


Gamification

- Learning streaks and daily goals
- Leaderboards and badges
- Community forums and live classes

Build It in Phases — Roadmap

Don't attempt to build all 40+ modules simultaneously. A phased approach lets you launch an MVP quickly, validate with real users, and add complexity only when the foundation is solid.



Phase 1 — MVP Checklist

Phase 1 gives you a fully functional LMS. Every item below is a prerequisite for launch. Don't move to Phase 2 until all of these are stable and tested in production.

1 Authentication

Registration, login, email verification, OAuth, password reset.

2 User & Admin Dashboards

Student learning view and admin control panel with KPIs.

3 User Management

Admin can view, create, edit, block, and delete users.

4 Course Management

Full CRUD for courses, sections, lessons, and categories.

5 Video Player & Security

Signed URLs, private storage, resume from last position.

6 Enrolment, Progress, Quiz & Certificate

Granular tracking, quiz scoring, and automatic certificate generation.



Database Architecture Overview

A well-designed schema is the foundation everything else is built on. Plan this before writing a single API endpoint. The tables below represent the minimum viable schema — you don't need all of them on day one.

Identity & Auth

users, roles, permissions, role_permissions, sessions, audit_logs

Courses & Content

courses, sections, lessons, lesson_resources, categories, subcategories, tags, course_tags

Learning & Assessment

enrollments, lesson_progress, quizzes, quiz_questions, quiz_attempts, assignments, certificates

Commerce & Marketing

orders, payments, refunds, coupons, coupon_usage, cart, wishlist, transactions, subscriptions

Community & CMS

reviews, questions, answers, notifications, support_tickets, banners, media, pages, blogs, settings

Instructors & Affiliates

instructors, instructor_applications, instructor_payouts, affiliate_users, referrals

Admin Sidebar Structure

A well-organised admin sidebar is critical for usability. Group related modules logically so that any admin or content manager can navigate to the right tool in under two clicks.

Core Modules

-  Dashboard
-  Users → Students, Instructors, Blocked
-  Courses → All, Pending, Published, Drafts, Categories, Quizzes
-  Instructors → Applications, Approved, Suspended, Payouts
-  Orders → All Orders, Payments, Refunds, Failed
-  Marketing → Coupons, Discounts, Campaigns, Affiliates

Platform & System

-  Website → Homepage, Banners, Media, Pages, Menus, Footer
-  Content → Reviews, Q&A, Blog, Reports
-  Communication → Notifications, Emails, Support Tickets
-  Analytics → Users, Courses, Revenue, Learning, Instructor
-  Certificates & Templates
-  Roles & Permissions
-  Audit Logs
-  Settings → General, Payment, Email, Storage, Security, Social

User Sidebar & Navigation

The student-facing sidebar should be clean, task-oriented, and focused on the learning journey. Every item should answer: "What does the student want to do right now?"

My Account Navigation

→ Dashboard

Welcome message, stats, continue learning, and recommendations.

→ My Learning

All courses, in-progress, completed, and wishlist.

→ Certificates

View, download, and share all earned certificates.

→ Orders & Payments

Purchase history, invoices, and payment methods.

→ Notifications & Messages

All alerts, replies, announcements, and direct messages.

→ Profile & Settings

Account details, password, privacy, and logout.



System Settings Module

A centralised settings panel means that configuration changes — from payment gateways to SMTP providers — never require a code deployment. Everything is managed by the super admin from the dashboard.

General

Website name, logo, favicon, contact email, phone, and address.

Payment

Gateway selection (Razorpay / Stripe / PayPal), currency, tax/GST rates, and invoice settings.

Email

SMTP provider, sender name, reply-to address, and template management.

Storage

Storage provider (S3, R2, GCS), file upload limits, and allowed MIME types.

Security

Password rules, max login attempts, session timeout, and 2FA enforcement for admins.

Social Media

Facebook, Instagram, YouTube, LinkedIn, and X/Twitter platform links.

Priority Feature Matrix

Not all features are created equal. Use this priority matrix to guide your sprint planning and ensure the most impactful features ship first.

Feature	Priority	Phase
Authentication & Role/Permission System	 Critical	Phase 1 — MVP
User & Admin Management	 Critical	Phase 1 — MVP
Course Management & Video Learning	 Critical	Phase 1 — MVP
Progress Tracking, Enrolment & Payments	 Critical	Phase 1–2
Quizzes & Certificates	 High	Phase 1 — MVP
Reviews, Wishlist, Coupons & Notifications	 High	Phase 3
Instructor System & Analytics	 High	Phase 3
Blog, Affiliate System & Mobile Apps	 Later	Phase 4–5
AI Tutor, Coding Sandbox & Gamification	 Later	Phase 5

Key Architectural Decision

The single most impactful architectural choice you'll make is how you design your authorisation system. Get this right from day one — retrofitting role-based permissions onto a binary `admin = true` system is enormously expensive.

✓ Design It This Way

A full permission hierarchy from the start:

- Super Admin → Admin → Content Manager
- Support Staff → Instructor → Student
- Each role maps to a set of `resource.action` permissions
- Every API route checks permission, not just role

✗ Not This Way

A flat binary model that breaks as soon as you add a third role:

- `isAdmin = true / false`
- No permission granularity
- Every new role requires major code changes
- Security gaps when roles overlap

⚠ A binary auth model is the #1 reason LMS rewrites happen 6 months after launch. Don't take the shortcut.

Next Steps — Turn Blueprint into Code

This blueprint is your strategic map. The next step is translating it into a technical specification that your development team (or you alone) can code from, sprint by sprint.



Define Frontend Pages & Routes

List every public page (`/courses`, `/course/:slug`), auth page, student page, and admin page with its route pattern.



Design API Endpoints

For each feature module, define REST (or GraphQL) endpoints: method, path, request body, response shape, and required permissions.



Set Up Project Structure

Define your monorepo or multi-repo layout, folder structure for frontend and backend, and CI/CD pipeline before the first commit.



Finalise Database Schema

Create the ERD for Phase 1 tables first. Define relationships, indexes, and constraints before writing migrations.

You're Ready to Build

You now have a complete, production-grade feature blueprint covering **40+ modules**, a **5-phase roadmap**, a **role-permission architecture**, a **full database schema**, and clear **priority guidance**.

Start with Phase 1

Authentication → Dashboards → Course Management → Video Player → Progress → Quiz → Certificate.

Add Phase 2 Next

Cart → Checkout → Payments → Orders → Coupons → Refunds → Invoices.

Scale to Phase 3+

Instructors, analytics, marketing, CMS, AI, mobile — layer in complexity only once the foundation is solid.

 The goal is a functioning, revenue-generating LMS as fast as possible. Ship Phase 1, get real users, then iterate. Every feature in this blueprint is worth building — but not all at once.